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RESEARCH LETTER

The greening of the Sahara during the mid-Holocene: results of an interactive atmosphere-biome model

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Abstract. An asynchronously coupled atmosphere-biome model was used to assess the biogeophysical interaction during the mid-Holocene, 6000 years before present. The model determines its own land surface conditions and can, therefore, predict quasi-equilibrium changes in global vegetation patterns. For the mid-Holocene, the coupled model simulated a northward shift of savanna up to 20°N into the Sahara which in turn amplified the response of the atmospheric circulation to changes in the Earth's orbit during the Holocene. Moreover, in the simulation, most of the western part of the Sahara appeared to be vegetated by xerophytic woods/shrub and warm grass, while the eastern part remained a desert. The simulated vegetation distribution agreed better with palaeobotanical reconstructions than results from the same model, but omitting vegetation-atmosphere interaction.

Key words. Atmosphere-biome interaction, biome, mid-Holocene, model, Sahara.

INTRODUCTION

Palaeoclimatological reconstructions using ancient lake sediments (Yu & Harrison, 1996), marine palynological data (Dupont, 1993), estimates of active sand dunes (Sarntheim, 1978), and archaeological evidence (Crowley & North, 1991) indicate that during the mid-Holocene, some 6000 years before present, the Sahel and Sahara regions were wetter than today. Moreover, it was found from fossil pollen (e.g. Jolly *et al.*, 1996) that the vegetation limit between the Sahara and the Sahel was at least as far north as 23°N. It has been hypothesized (e.g. Kutzbach & Guetter, 1986) that the wetter climatic conditions were probably caused by changes in the Earth's orbit. 6000 years ago, the perihelion was in mid-September, the tilt of the Earth's axis was 0.7° stronger than today, and the eccentricity of the Earth's orbit around the sun was a little bit larger (0.187 instead of 0.167 today). Numerical simulations using general circulation models (GCMs) of the atmosphere revealed that these changes in the Earth's orbit led to an increased solar radiation in the Northern Hemisphere summer which amplified the African and Indian summer monsoon, thereby increasing the moisture transport into North Africa (Kutzbach & Guetter, 1986; Lorenz *et al.*, 1996).

Most simulations, however, underestimate the climate change. For example, simulations within the framework of the Palaeoclimatic Modelling Intercomparison Project (PMIP) did not show a substantial wetting north of approximately 20°N (see Yu & Harrison, 1996). Moreover, when analysing the simulation by Lorenz *et al.*, 1996—in the following referred to as Exp-0—in terms of global biome patterns using the BIOME model of Prentice *et al.*, 1992 (see Fig. 1), the vegetation patterns agree, by and large, with vegetation patterns from simulations of present-day climate with the same model (Fig. 2). Exceptions are the boreal biomes in Europe which moved in general one grid box, i.e. approximately 2.8°, corresponding to some 300 km, farther north. Moreover savanna* is found at the southern rim of the Sahara in Exp-0, instead of the xerophytic woods/shrub located there in the simulation of present-day climate. Only a very few

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* The terminology of biomes strictly follows that of the BIOME model by Prentice *et al.* (1992).

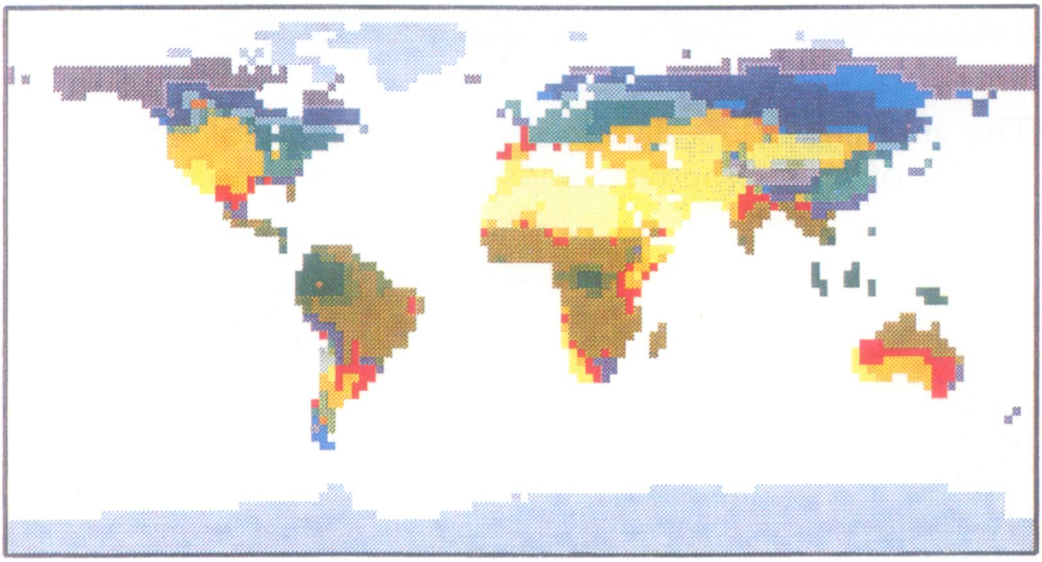

Fig. 1. Vegetation patterns computed from a simulation of the mid-Holocene climate, 6000 years ago, by Lorenz *et al.* (1996) using the Hamburg atmospheric general circulation model ECHAM (Exp-0). In this experiment, only the Earth's orbit parameters are changed while the other boundary conditions, ocean temperatures and land surface, remain unchanged at present-day values.

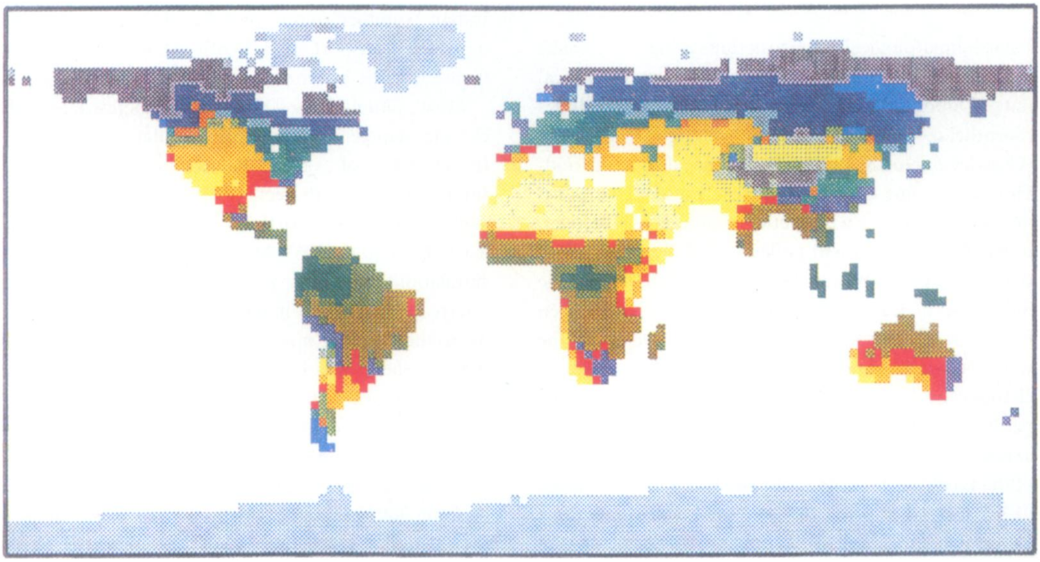

Fig. 2. Vegetation patterns computed from a simulation of present-day climate (with the same model whose results are depicted in Fig. 1).

Key to Figs 1, 2 & 4

grid boxes in today's Sahara are covered by warm grass in Exp-0. Hence there is a mismatch with palaeoclimatic reconstructions.

Presumably, this weak response to orbital forcing can be attributed to the prescribed land surface conditions. In the PMIP simulations, the parameter of the Earth's orbit and the atmospheric concentration of greenhouse gases were changed to mid-Holocene values, while the other boundary conditions, sea surface temperature (SST) and land-surface characteristics, remained unmodified at present-day values. Sensitivity studies using GCMs with prescribed, but modified land-surface conditions (e.g. Street-Perrott *et al.*, 1990; Bonan, Pollard & Thompson, 1992, Foley *et al.*, 1994) have suggested that large positive feedbacks between climate and vegetation may have taken place in the late Quaternary. Recently, Kutzbach *et al.* (1996) demonstrated that changes in vegetation and soil tend to enhance the climate response to orbital forcing. They found that changes in orbital parameters alone increased summer precipitation by 12% between 15°N and 22°N. Replacing desert with grassland, and desert soil with more loamy soil, further enhanced the summer precipitation by 28%. When the simulated climate changes were applied to the BIOME model, then the area of the Sahara became smaller by 11% owing to orbital forcing alone, and by 20% owing to the combined influence of orbital forcing and the prescribed vegetation and soil changes. Although the simulations with modified land-surface yielded an increased response of the climate to orbital forcing, the resulting climate change was still too small compared with what is shown by the palaeodata.

Here, we reassess the simulation of Lorenz *et al.* (1996). However, instead of prescribing land-surface conditions, we use a coupled atmosphere-vegetation model. Such a model-system determines its own land-surface characteristics and can, therefore, predict quasi-equilibrium changes in global vegetation patterns (e.g. Henderson-Sellers, 1993; Claussen, 1994). When a coupled atmosphere-vegetation model was applied to present-day climate conditions, interesting consequences concerning the bio-geophysical feedback in the Sahara–Sudan region were found (Claussen, 1997a). The simulations revealed that two solutions of the atmosphere-vegetation system can exist. Depending on the initial land surface conditions, the present-day Sahara is retrieved as well as a greening of the south-western part of it mainly by savanna and xerophytic woods/shrub. Therefore, it seems worthwhile to investigate whether the atmosphere-vegetation model yields a 'green solution' also for the mid-Holocene period.

METHODS

The atmospheric model

The GCM used in this study, the atmospheric model ECHAM (version 3.2), was developed at the Max-Planck-Institut für Meteorologie in Hamburg (see Roeckner *et al.*, 1992). ECHAM is run at T42 resolution, hence the Gaussian grid has a resolution of some $2.8° \times 2.8°$ longitude/latitude. ECHAM is able to simulate most aspects of the observed time-mean circulation and its intraseasonal variability with remarkable accuracy. However, during the respective summer season, there is too much precipitation over South Africa and Australia and off the west coast of Central America, while rainfall over India is underestimated during the summer monsoon season. Moreover, there is a lack of precipitation over the continents in the Northern Hemisphere in summer. On the other hand, the global vegetation patterns diagnosed from ECHAM simulations of present-day climate by using the BIOME model of Prentice *et al.* (1992) yielded fair agreement with observations—except for some deviations in the regions mentioned above (Claussen & Esch, 1994).

The biome model

Biomes are computed by using the BIOME model (version 1.0) developed by Prentice *et al.* (1992). This model is based on physiological considerations rather than on mere correlations between climate and biomes as they exist today. Therefore, the BIOME model is quite suitable as a tool to assess changes in natural vegetation in response to changes in climate. However, it is important to note that the BIOME model does not simulate the transient behaviour of vegetation. At best, it provides constraints within which plant community dynamics could operate.

In the BIOME model, fourteen plant functional types are assigned climate tolerances in terms of amplitude and seasonality of climate variables such as coldest monthly mean temperature, yearly temperature sums, and the ratio of actual to equilibrium evapotranspiration. Competition between different plant types is treated indirectly by the application of a dominance hierarchy which effectively excludes certain types of plants from a site based on the presence of other types rather than because of climate. Finally, the biomes are defined as combinations of dominant types.

Coupling of the atmospheric with the biome model

ECHAM is asynchronously coupled with the BIOME model: from climate variables computed by ECHAM, biome groups are diagnosed by using the BIOME model. The effect of vegetation on the near-surface energy, moisture, and momentum fluxes computed in the atmospheric model is parameterized in terms of surface parameters. These parameters include background albedo, roughness length, vegetation ratio and forest ratio (relative area of a grid cell covered by vegetation and forest, respectively), and leaf area index (area of leaves covering 1 m^2 surface area). Allocation of surface parameters to biomes is described in detail in Claussen (1994, 1997a). In Claussen (1997a), the biome 'sand desert' which is not included in the original BIOME model was introduced to take into account the large differences between albedo values of various types of desert. Only in the Sahara and Arabian deserts are large patches of high albedo—in some place up to 0.42—observed by satellite (e.g. Ramanathan *et al.*, 1989). Therefore, whenever the BIOME model predicts 'hot desert' in these regions, 'sand desert' was chosen. Hence 'sand desert' is not defined in ecological but in meteorological terms.

Using the new surface parameters, the simulation with ECHAM is continued for some model years. Based on earlier investigations (Claussen, 1994), six years are chosen from which the first year is omitted as an adjustment period for soil-water transports. From the new simulation biomes are diagnosed again and so on. The sequence of allocation of surface parameters to biome groups, integration with ECHAM, and computation of biomes using the BIOME model is referred to as 'iteration'. Several interations are performed until the biome patterns and the state of the atmosphere no longer reveal any significant trend.

RESULTS

Vegetation response

Using the output of Exp-0, i.e. of the atmosphere-only model ECHAM, and prescribing the vegetation

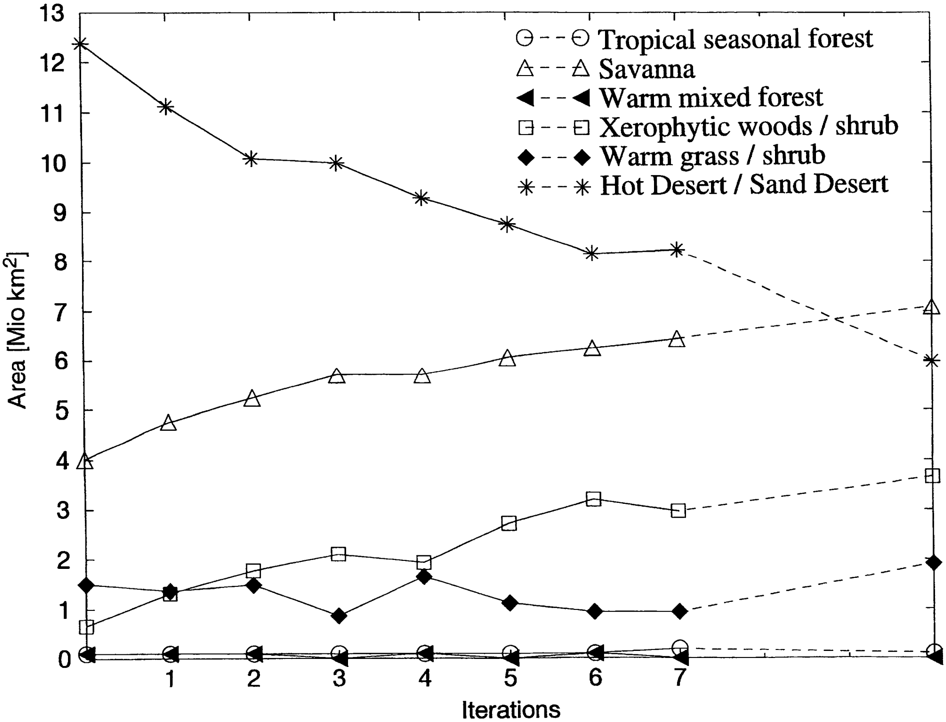

Fig. 3. Area (in 10^6 km^2) of vegetation zones in North Africa (15.5°W–52.0°E and 8.5°N–36.6°N) for each iteration in Exp-1. This experiment was started with the vegetation patterns computed from Exp-0 (see Fig. 1). The atmosphere-vegetation model drifts towards the solution obtained in Exp-2 (see Fig. 4). The iteration steps do not reflect real time intervals.

distribution shown in Fig. 1 as the initial land-surface condition, the ECHAM-BIOME model starts to drift. It appears that vegetation intrudes into the former desert, i.e. after each iteration, the Sahara desert shrinks more and more. Mainly the biomes 'savanna' and 'xerophytic woods/shrub' grow at the expense of 'desert' (see Fig. 3). Unfortunately, Exp-1 was interrupted by severe computer problems.

A new numerical experiment, Exp-2, was started. In Exp-2, the initial land-surface conditions were chosen to be the same as in Exp-1, with the exception that the Sahara desert was replaced by rain forest. These new, rather unrealistic initial conditions were chosen for the following reasons. First, we did not want to start with the best reconstruction of paleovegetation currently available. Instead, the coupled model should find the vegetation patterns itself. Second, previous experiments (Claussen, 1994, 1997a) revealed that the atmosphere-vegetation system can behave as an intransitive system. Depending on initial conditions of high or low albedo in the Sahara the coupled model yields either a desert or a vegetated Sahara. It was found that the changes in albedo were more important than changes in other surface parameters. Prescribing the albedo of sand desert, but the vegetation density (vegetation ratio and leaf are index) of grassland, brought forth very much the same results as prescribing both the albedo and vegetation density of sand desert*. Furthermore, recent simulations (Claussen, 1997b; Kubatzki, 1996) showed that if the system is intransitive, then it has only two solutions. The ECHAM-BIOME model did not attain more than two solutions, depending on the type of biomes or the number of grid cells changed from sand desert to any other biome. Any reduction in the albedo from the value allocated to sand desert to values allocated to other biomes resulted in an encroachment

* Vegetation ratio and leaf area index of sand desert were assumed to be zero.

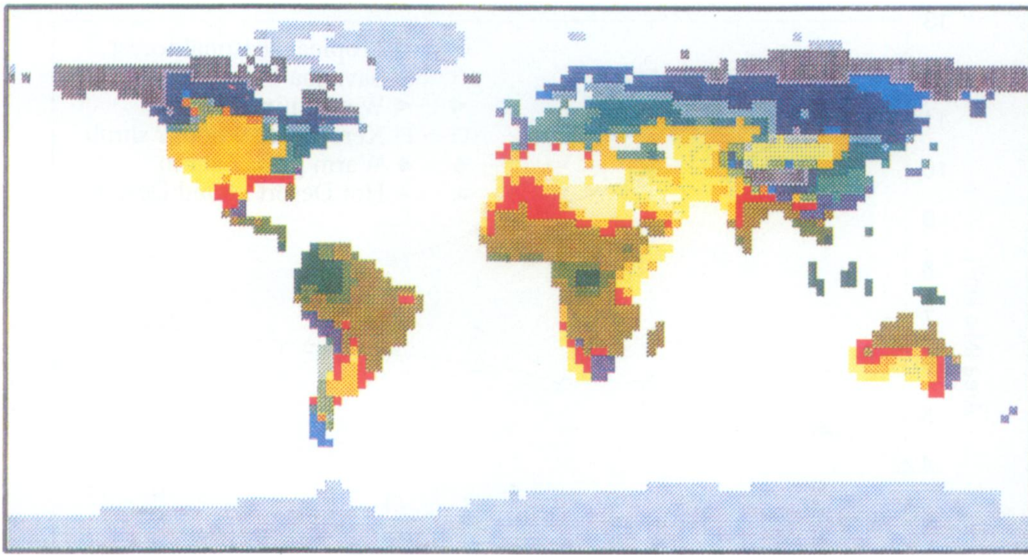

Fig. 4. Same as Fig. 1, except that climate and resulting vegetation patterns are computed from the interactively coupled ECHAM-BIOME model.

of xerophytic woods/shrub into the Sahara. Third, the ECHAM-BIOME model found its solution more quickly when initialized with surface conditions closer to the solution. Any large-scale, strong change in albedo would 'kick' the coupled model to one or the other equilibrium solution.

Starting from the new initial conditions, the ECHAM-BIOME model 'jumps' to an equilibrium. In Exp-2, only three iterations were performed, since vegetation maps from all iterations resemble each other and there is no significant trend in vegetation zones. (Due to internal climate variability, different numerical realizations of the same climate state do not yield the same, but slightly different, vegetation patterns, see Claussen, 1996.) Fig. 4 depicts the vegetation distribution of the third iteration of Exp-2. In a third experiment, grassland was initially prescribed instead of rain forest in the Sahara. Again, the ECHAM-BIOME model immediately attained a solution similar to that shown in Fig. 4, thereby corroborating the results summarized in the previous paragraph.

The main difference between the vegetation distributions resulting from Exp-0 and from Exp-2 is obviously the northward shift of savanna of some 600 km up to approximately 20°N in the western part of the Sahara. North of the savanna, xerophytic woods/shrub and warm grass are predicted, which cover the western part of the Sahara up to the Mediterranean Sea. Only the eastern part of the Sahara remains a desert. The vegetation pattern of Exp-2 closely resembles that of the 7th iteration of Exp-1, therefore, it is not shown here.

Atmospheric response

What causes the vegetation to shift? In Table 1, the climate near the ground is depicted on average over the interior of the Sahara (10°W–30°E and 15°N–30°N) during the summer months (June, July, August) of the mid-Holocene. For the winter season negligibly small differences between experiments were found. Table 1 shows that the net shortwave radiation at the surface is, within 1 W/m^2, the same for the desert in Exp-0 and for the green Sahara in Exp-2—despite a much higher albedo of the desert. This is caused by the increase in cloudiness above the vegetated Sahara. Increased cloudiness is also responsible for an increase in atmospheric (thermal) radiation. In total, the vegetated surface in Exp-2 receives some 30 W/m^2 more energy due to radiation than the desert in Exp-0. This excess energy is mainly used for evaporation. The Bowen ratio (the ratio of sensible to latent heat flux) is almost 3 in Exp-0 and approximately 0.5 in Exp-2. These values are typical

Table 1. Surface-layer climate on average over the Sahara (10°W–30°E and 15°N–30°N) for summer months of June, July, August, 6000 years ago. Exp-0 refers to a climate simulation in which the land surface was prescribed as it is today. Exp-2 refers to a simulation with a coupled atmosphere-vegetation model. Negative values indicate upward fluxes.

	Exp-0	Exp-2
Albedo	0.34	0.22
Incident shortwave radiation	339 W/m^2	284 W/m^2
Outgoing shortwave radiation	–117 W/m^2	–62 W/m^2
Net shortwave radiation	222 W/m^2	222 W/m^2
Atmospheric radiation	393 W/m^2	408 W/m^2
Outgoing thermal radiation	–513 W/m^2	–498 W/m^2
Sensible heat flux	–62 W/m^2	–47 W/m^2
Latent heat flux	–21 W/m^2	–82 W/m^2
Evapotranspiration	22 mm/month	85 mm/month
Precipitation	30 mm/month	129 mm/month
Soil moisture	19%	41%
Surface temperature	34.6°C	31.5°C
Cloud cover	0.23	0.47

for semi-arid and more densely vegetated, moister regions, respectively.

Since the vegetated surface evaporates more than the desert, it becomes colder. This would imply a weaker gradient in surface temperatures between land and the Atlantic ocean, leading to a decrease in the African Monsoon (Meehl, 1994). However, the essential point is that the total energy (sensible and latent heat) put into the atmosphere is increased. Thereby, the vertical gradient of moist static energy is increased over the vegetated surface. It is the vertical gradient of moist static energy which determines the convective precipitation (Charney, 1975; Charney, Stone & Quirk, 1975). Changes in precipitation, in turn, feed back on vegetation.

Comparing the values of precipitation P minus evaporation E in Table 1, it is evident that $P-E$ in Exp-2 is much larger than in Exp-0. This indicates that not all of the precipitation is locally produced. Some of it is advected. Inspection of wind climatologies (not shown here) revealed that the moisture is taken from the tropical Atlantic by a stronger African monsoon. This process is the same as that observed in simulations with the ECHAM-BIOME system for present-day conditions and conditions prevailing during the last glacial maximum (LGM). Also for present-day and LGM climate, an increase in vegetated area in North Africa leads to a positive feedback with the African summer monsoon (Claussen 1997a,b; Kubatizki, 1996). The reason for this is the increase in convection (vertical motion) above the vegetated area which has to be compensated by a stronger lateral flow.

DISCUSSION AND CONCLUSION

Geological reconstructions indicate that the climate in the Sahara during the mid-Holocene (6000 y BP) was much wetter than today, and vegetation was found far north of the present desert border. Using an interactively coupled atmosphere-vegetation model, this shift of vegetation has been recaptured. On the other hand, when we used an atmosphere-only model and prescribed present-day land-surface conditions, the enhancement of simulated precipitation was insufficient to diagnose an encroachment of xerophytic woods/shrub into the Sahara north of approximately 20°N. The difference between the two types of simulation was explained in the framework of Charney's (1975) theory of a self-induction of deserts through albedo enhancement in the Sahel.

The weak shift of vegetation in the atmosphere-only model is not only observed in the ECHAM model. Other atmospheric circulation models yield the same qualitative result (see Yu & Harrison, 1996). Also Kutzbach *et al.* (1996), who prescribed grassland instead of desert in the Sahara, found a rather weak response in comparison with our simulations. Presumably the difference between Kutzbach *et al.*'s and our simulation can be attributed to the difference in albedo which turned out to be the most important factor concerning the biogeophysical feedback. Kutzbach *et al.* state that the change from desert to grassland decreases the surface albedo slightly. In our study, the albedo was reduced from 0.34 to 0.22 in the region 10°W–30°E, 15°N–30°N (the albedo value allocated to sand desert was 0.35, the value for warm grass 0.2). This reduction is a rather large one, but it is in agreement with satellite data of today's differences between Sahara desert and steppe (e.g. Ramanathan *et al.*, 1989). For the Sudan region (15°N–22°N, 0°E–50°E), Kutzbach *et al.* reported a reduction from 0.25 to 0.22. In our study we found a stronger reduction of 0.27 in Exp-0 to 0.18 in Exp-2. Hence this point needs further consideration.

There are too few data or even proxy data to validate fully the ECHAM-BIOME model. By and large, the atmosphere-vegetation model agrees with the reconstruction of Jolly *et al.* (1996) who find pollen indicative of xerophytic savanna up to 23°N in the western part of the Sahara and pollen indicative of warm grass in the eastern part. Reconstructions of Dupont (1995) indicated a less extensive northward spread of vegetation than found by Jolly *et al.* (1996). However, Jolly *et al.*'s reconstruction may be more trustworthy, because Dupont's result was based on an indirect estimate: it was calculated by analysis of pollen which were deposited in marine sediments off the North African coast and which were presumably transported from the continent into the Atlantic by the trade winds and by the African easterly jet. Frenzel, Pesci & Velicho (1992) estimated that the minimal deviation of annual precipitation from present-day values amounts to 200–300 mm near the Tropic of Cancer. This estimate agrees better with the results of Exp-2 than with those of Exp-0. Exp-0 yielded some 0–100 mm. Exp-2 generated approximately 350 mm.

Despite overall agreement, the following question remains. Exp-2 (as well as Exp-1) reveals a strong east-west gradient of moisture. For example, at approximately 20°N, Exp-2 yields up to 800 mm more annual precipiation than the simulation of present-day climate in the western part and some 200 mm more in the eastern part of the Sahara. Evidence of a wet western part and a drier eastern part during the mid-Holocene period of more intense monsoon activity is consistent with model simulations of Rodwell & Hoskins (1996). On the other hand, lake status data (Yu & Harrison, 1996) indicate wet conditions in today's Libyan and Egyptian deserts, which conflicts with the dry conditions produced in the simulation. This difference points at a problem in the atmospheric model ECHAM. ECHAM, in its version 3, does not have a lake model. Hence lakes cannot be generated, nor can lakes which are smaller than a grid cell in ECHAM affect near-surface energy and moisture fluxes. Because $P-E$ amounts to approximately 10–50 mm/month on an annual average in Exp-2—in contrast to Exp-0 in which $P-E$ vanishes in these regions—surface run-off could have formed lakes.

In conclusion, palaeoclimatic reconstructions indicate that the simulated climate and vegetation pattern of the mid-Holocene from the ECHAM-BIOME model are 'within the right ball park'. The atmosphere-only model (ECHAM, as well as others) definitely underestimates the difference between present-day and mid-Holocene climate. Assuming that mid-Holocene SST were not significantly different from present-day values (which is in line with geological evidence, e.g. Crowley & North, 1991) we therefore suggest that changes in the Earth's orbit alone are insufficient to explain the mid-Holocene climatic optimum. It appears that biogeophysical feedbacks in the African Monsoon regime tend to amplify this climate change leading to a northward shift of vegetation into the Sahara which in turn leads to a moister climate in this region.

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